

Reviewer Pack — Young Tableaux Ratio Exponential Lower Bound on Permanent Ci...

Entry 37932008c65e · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-09 02:52:47 UTC

Young Tableaux Ratio Exponential Lower Bound on Permanent Circuit Size

Entry ID: 37932008c65e **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-09 02:52:47 UTC

1. Conjecture

Field A (mathematical branch): Representation Theory of Symmetric Groups **Field B** (complexity object): Boolean Circuit Size

Statement:

For random $n \times n$ matrices, the ratio of standard Young tableaux counts for shape $(n, n, \dots, n)$ (n times) to shape $(n, n-1, \dots, 1)$ is $\Omega(2^n)$. This implies permanent requires $\Omega(2^n)$ size circuits for $n \times n$ matrices.

Rationale (proposer's reasoning):

Young tableaux counts encode symmetry-breaking patterns in tensor decompositions. The exponential gap in tableau ratios reflects the inherent asymmetry between permanent's fully symmetric structure and determinant's staircase symmetry, suggesting computational hardness.

Taxonomy category: GCT_DET_PERM (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 7579eafc27b08629

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.8s

5.1 Generated Python source

```
import random
import math
from fractions import Fraction

def factorial(n):
    if n == 0:
        return 1
    result = 1
    for i in range(2, n + 1):
        result *= i
    return result

def hook_length_formula(matrix, shape):
    n = len(shape)
    numerator = 1
    denominator = 1
    for row in range(n):
        for col in range(n):
            hook_length = (n - row) + (n - col) - 1
            numerator *= factorial(hook_length)
            denominator *= matrix[row][col]
    return Fraction(numerator, denominator)

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.choice([5, 10, 15, 20, 30, 40])
    instances_tested = 0
    conjecture_holds = True
    counterexample = ""

    for _ in range(30):
        matrix = [[random.randint(1, n) for _ in range(n)] for _ in range(n)]
        rectangular_shape = (n,) * n
        staircase_shape = tuple(range(n, 0, -1))

        rect_count = hook_length_formula(matrix, rectangular_shape)
        stair_count = hook_length_formula(matrix, staircase_shape)

        ratio = rect_count / stair_count
```

```

    if ratio < 2**n:
        conjecture_holds = False
        counterexample = f"Matrix with n={n} failed. Ratio: {ratio}"
        break

    instances_tested += 1

    return {
        "metric_name": "Young Tableaux Ratio",
        "metric_value": 2**n,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i + 3 for i in range(5, 8)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mean_ratio = sum(r["metric_value"] for r in results) / len(results)
    support_fraction = sum(r["conjecture_holds"] for r in results) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_ratio} std=0.0 support_fraction=1.0")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_ratio} std=0.0 support_fraction={support_fraction}")
    else:
        first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
        print(f"RESULT: FALSIFIED counterexample=\"{result['counterexample']}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

ith n=5 failed. Ratio: 1'}
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 32768, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'Matrix with n=5 failed. Ratio: 1'}
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 32768, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'Matrix with n=5 failed. Ratio: 1'}
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 1099511627776, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'Matrix with n=5 failed. Ratio: 1'}
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 1073741824, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'Matrix with n=5 failed. Ratio: 1'}
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 1024, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'Matrix with n=5 failed. Ratio: 1'}
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 1073741824, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'Matrix with n=5 failed. Ratio: 1'}
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 1024, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'Matrix with n=5 failed. Ratio: 1'}
TRIAL: {'metric_name': 'Young Tableaux Ratio', 'metric_value': 1024, 'instances_tested': 0, 'conjecture_holds': False, 'counterexample': 'Matrix with n=5 failed. Ratio: 1'}

```

--STDERR--

Traceback (most recent call last):

File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_90d1a1b1.py", line 87, in <module>

```
first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
```

KeyError: 'seed'

Critic verdict: CHALLENGE

skipped: test crashed before producing data

Verdict: INCONCLUSIVE

```
parse_fail | next: NONE
```

Total LLM calls: 8

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 286290 ms total latency. Provider mix: {'ollama remote': 8}

12. Reproducibility

To reproduce this cycle:

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)